

Continuous and Interpretable Empathic Behaviors for Embodied Intelligent Virtual Agents

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Figure 1: Example interaction of the proposed empathic intelligent virtual agent. The agent conveys empathy by continuously modulating eye contact, nodding, mimicry, body proximity, and facial expression, guided by an internal empathy state.

ABSTRACT

We present an embodied intelligent virtual agent (IVA) for empathic human-agent communication in immersive extended reality (XR). The agent supports low-latency streaming conversation using a commercial artificial intelligence (AI) service and synchronizes speech with viseme-driven lip sync and facial dynamics. During communication, the agent's empathy state and nonverbal behaviors are computed from user interaction data, including gaze, facial expressions, gestures, and vocal features. The empathy state integrates understanding-oriented and affect-oriented components, enabling the agent to express empathic behavior through eye contact, nodding, mimicry, body proximity, and facial expression in a continuous and interpretable manner. Our demonstration highlights the potential of grounding empathic IVA behavior in empirical observations of externally expressed empathy in dyadic XR communications.

Index Terms: Intelligent virtual agent, empathy, external expression, extended reality.

1 INTRODUCTION

Recent IVA frameworks have shown that commercial cloud-based AI services can be integrated with embodied agents in immersive XR to support human-AI communication and nonverbal behaviors [2]. However, many existing systems rely on multi-stage speech pipelines, which can introduce noticeable turn gaps. Interim idle or backchannel animations may partially mask these delays, but they often remain apparent during interaction. Moreover, agent behaviors such as facial expressions and body motions are com-

monly triggered at turn boundaries and treated as discrete animation choices aligned with model outputs.

Our demo instead focuses on transferring empathic behaviors observed in human-human virtual reality (VR) studies into an embodied IVA. Prior work quantified externally expressed empathy in dyadic VR communication [1] and reported associations between higher empathy and eye contact dynamics, stability of gaze and head motion, mimicry across facial expressions, gestures, and vocal features, as well as stable interpersonal proximity. These findings motivate a clear design goal for our system. Empathy-related cues should be expressed continuously during human-agent interaction, rather than appearing only as occasional, turn-level effects.

We present an empathic behavior layer that maintains an explicit agent empathy state and continuously modulates eye contact, nodding, mimicry, body proximity, and facial expression during interaction. The design is intended to remain interpretable and practical for real-time XR human-AI interaction systems.

2 SYSTEM OVERVIEW

Low-latency conversational IVA pipeline. To support low-latency conversation, we built a streaming speech pipeline using the commercial OpenAI realtime-mini API through a WebRTC session. The pipeline enables end-to-end conversational interaction with measured latency in the range of 500–1300 ms and allows large language model (LLM) outputs to be structured via prompting to emit explicit affective and empathy-related states. Agent audio is lip-synced using OVRlipSync API¹ with viseme-based mouth blend-shapes, inheriting the limitations of viseme-level articulation, with empathic facial dynamics layered on top. We implemented the system in Unity 6000.0.56f1 with an avatar authored in Character Creator 4 (CC4) and imported into a High Definition Render Pipeline (HDRP) project for high-quality avatar rendering.

Agent empathy state. User interaction data are captured using Meta Quest Pro and the Meta XR Movement SDK 81.0.0. We collect gaze, facial expression weights, full-body motion, and voice

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¹<https://developers.meta.com/horizon/documentation/unity/audio-ovrlipsync-unity/>

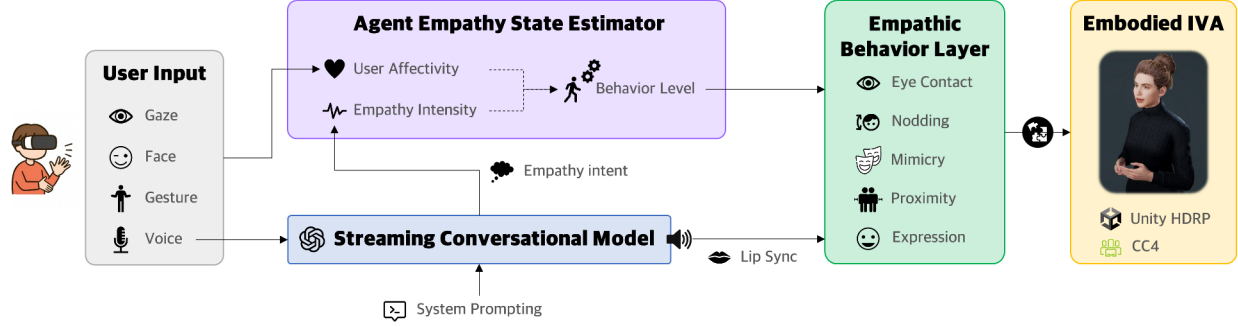


Figure 2: System overview of the proposed empathic intelligent virtual agent. User input modalities (gaze, facial expression, gesture, and voice) are used to estimate the agent’s empathy state, represented by user affectivity and empathy intensity. A streaming conversational model provides spoken responses and optional empathy intent. The estimated state drives an empathic behavior layer that modulates eye contact, nodding, mimicry, body proximity, and facial expression to synthesize continuous empathic behavior in the embodied IVA.

features, including short-term energy and speaking activity. Based on findings that different external cues relate to cognitive and emotional empathy in dyadic VR interactions [1], we represent the agent’s empathy state using two scalar variables, $E(t)$ and $w(t)$. $E(t)$ represents empathy intensity and scales the overall strength of the agent’s empathic behaviors, optionally biased by model-side empathy intent. $w(t)$ represents user affectivity and controls the mixture between two components, $\pi_{cog}(t)$ and $\pi_{aff}(t)$, which correspond to cognitive- and emotional-empathy-oriented behavior patterns, respectively:

$$\mathbf{B}(t) = \mathbf{B}_0(t) + E(t) \cdot ((1 - w(t)) \cdot \pi_{cog}(t) + w(t) \cdot \pi_{aff}(t)). \quad (1)$$

Here, $\mathbf{B}_0(t)$ denotes baseline behavior parameters produced by streaming lip sync and a small set of authored base animations. The weight $w(t)$ is derived from user affect cues such as facial affect intensity and vocal dynamics. When user affect is more salient, $w(t)$ increases, placing greater emphasis on the affect-oriented component. To improve robustness, input features are smoothed and $E(t)$ and $w(t)$ evolve more slowly than speech and lip-sync signals. Optionally, the language model is prompted to emit a soft empathy intent signal, which nudges $E(t)$ and $w(t)$ while preserving continuous behavior synthesis.

Synthesizing empathic behaviors. The empathic behavior layer is composited on top of the baseline animation and lip-sync outputs. It continuously modulates five behavior channels, grounded in cue patterns reported in prior analyses [1]:

- **Eye contact** is produced as brief gaze pulses with controlled gaze aversion. With stronger empathic expression, pulses occur more frequently. When user affect is more salient, pulses tend to last longer, consistent with observations that eye contact duration is informative for emotional empathy.
- **Nodding** is generated as backchannel cues whose frequency and timing are modulated by the agent’s empathy state to reflect both cognitive and emotional empathy.
- **Mimicry** is implemented as delayed and attenuated replay of selected cues. For facial cues, understanding-oriented behavior emphasizes subtle mouth-related responses during user speech, whereas affect-oriented behavior emphasizes a small set of expressions when user affect is salient. Gesture cues include restrained alignment of head and body motion. Voice cues optionally align short-term intensity trends, consistent with empathy-related vocal coordination patterns.
- **Body proximity** cues maintain an interaction-ready posture while respecting interpersonal distance. The proximity policy prioritizes stable approach behavior and suppresses oscillation, consistent with reduced proximity variability.

- **Facial expression** derives affect-relevant facial actions from the conversational model’s affective intent, which reflects the user’s affectivity. The resulting expression remains stable and interpretable during interaction.

3 DEMONSTRATION

The demo is presented on a desktop PC equipped with an RTX 5090 GPU and a Meta Quest Pro headset. Only the user’s speech audio is streamed to the cloud-based conversational model, while all other signals are processed locally and discarded after the session. Within a single session, the interface supports switching between baseline and empathy-enabled conditions for direct qualitative comparison of agent behavior. A spectator monitor provides a view of the interaction and can additionally display simple indicators of empathy intensity, affective salience, and active behavior channels.

4 CONCLUSION

This demo presents an embodied IVA for empathic communication in immersive XR. The system combines a low-latency streaming conversational pipeline with an empathic behavior layer grounded in empirical observations of externally expressed empathy in dyadic VR interactions. The agent’s empathy state continuously guides the agent’s behavior, such as eye contact, nodding, mimicry, facial expression, and body proximity during interaction. Future work will explore improved streaming speech interfaces and more natural facial and motion synthesis. Another direction includes controlled user studies to assess perceived empathy and modeling the temporal evolution of empathic agent–user relationships.

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